Spatial Data Analysis in R

Data Science II

Tec de Monterrey

August – December 2026

bgpopescu.net/teaching/gis2

Course Information

Course Title: Data Science II
Module: Spatial Data Analysis in R — 5 sessions
Term: August – December 2026
Schedule: 09:10 – 10:50 hrs
Location: Cedetec, Salón 210

Instructor

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Office Hours: By appointment

Module Description

This is the spatial module of Data Science II. Over five sessions it takes you from never having opened a shapefile to producing a complete accessibility analysis, pulling a continuous surface into your own table, and publishing the result as a report you can send to someone.

The other modules of this course cover large language models, APIs, web scraping, and SQL. This one covers the thing none of them touch: **data that has a location**. That is not a niche skill — municipio-level indicators, service coverage, census geography, and crime statistics are all spatial, and treating them as ordinary tables throws away most of what they can tell you.

Everything is done in **R**, using the `sf` and `terra` packages, and written up in **Quarto**. You have both already; we repurpose them.

Learning Outcomes

Upon completion of this module, students will be able to:

1. **Explain what a coordinate reference system is** and diagnose the errors that follow from getting it wrong.

2. **Read, inspect, and write spatial data** in the common vector formats, and build a spatial layer from an ordinary table of coordinates.
3. **Join data to geography** — by administrative code and by location — and know which to trust when the two disagree.
4. **Produce a defensible thematic map**: choose a rate rather than a count, classify it deliberately, and colour it so the reader is not misled.
5. **Measure**: areas, distances, nearest features, buffers, and overlays.
6. **Use raster data**, and extract a continuous surface into a column of your table.
7. **Publish** the result as a self-contained Quarto document.

What You Need

A laptop with at least 8 GB of RAM, and:

- **R 4.4 or later** and **RStudio** — posit.co/download/rstudio-desktop
- **Quarto** — quarto.org
- The packages installed in Session 1:

```
install.packages(c("sf", "terra", "tidyterra", "exactextractr",
                   "ggplot2", "dplyr", "tidyverse", "patchwork",
                   "classInt", "viridis", "leaflet",
                   "spData", "cowplot", "ggspatial"))
```

Assessment

This module does not carry its own grade. It feeds the course **reto**.

Module Content

Session	Topic
1	Quarto for Spatial Reporting
2	Coordinates, CRS, and Your First Map
3	`sf` Essentials, Choropleths, and Interactive Maps
4	Spatial Joins, Measurement, and Buffers
5	Rasters: Continuous Surfaces and Zonal Statistics
Opt.	Creating a Website with GitHub Pages and Quarto

Session dates are set by the course calendar; see Canvas.

Books and References

- Lovelace, Robin, Jakub Nowosad, and Jannes Muenchow. *Geocomputation with R.* r.geocompx.org — the standard reference, free online.
- Pebesma, Edzer, and Roger Bivand. *Spatial Data Science: With Applications in R.* r-spatial.org/book — by the author of `sf`.
- Gimond, Manuel. *Intro to GIS and Spatial Analysis.* mgimond.github.io/Spatial — strongest on the conceptual side.
- Hijmans, Robert. *Spatial Data Science with R and terra.* rspatial.org — by the author of `terra`.

Academic Integrity

The use of artificial intelligence is permitted within the university's ethical guidelines and applicable law, under one condition: every submission includes an appendix stating which AI tools were used, for what purpose, and with which prompts. Undisclosed AI use is academic dishonesty.

Any act of academic dishonesty will result in a failing grade on the work in question, and will be reported to the academic administration.

Attendance

Attendance follows the course policy set by the coordinating instructors. Students with a justified reason for missing a session should email the instructor beforehand.

Students with Special Needs

The university does not discriminate based on disability. Students with approved accommodations must inform their instructors at the beginning of the term. Please see the institutional website for the complete policy.

1 Quarto for Spatial Reporting

Lecture

Open the Slides

Reading

Lovelace, Nowosad, and Muenchow. *Geocomputation with R.* Ch. 2, Geographic data in R. (link)

Quarto. *HTML Basics.* (link)

2 Coordinates, CRS, and Your First Map

Lecture

Open the Slides

Reading

Gimond, Manuel. *Intro to GIS and Spatial Analysis.* Ch. 9, Coordinate Systems. (link)

Lovelace, Nowosad, and Muenchow. *Geocomputation with R.* Ch. 7, Reprojecting geographic data. (link)

Gimond, Manuel. *Intro to GIS and Spatial Analysis.* Ch. 2, Feature Representation. (link)

3 `sf` Essentials, Choropleths, and Interactive Maps

Lecture

Open the Slides

Reading

Gimond, Manuel. *Intro to GIS and Spatial Analysis.* Ch. 5, Statistical Maps. (link)

Gimond, Manuel. *Intro to GIS and Spatial Analysis.* Ch. 6, Pitfalls to Avoid. (link)

Gimond, Manuel. *Intro to GIS and Spatial Analysis.* Ch. 7, Good Map Making Tips. (link)

Lovelace, Nowosad, and Muenchow. *Geocomputation with R.* Ch. 9, Making maps with R. (link)

Lovelace, Nowosad, and Muenchow. *Geocomputation with R.* Ch. 8, Geographic data I/O. (link)

4 Spatial Joins, Measurement, and Buffers

Lecture

Open the Slides

Practice Quiz

Ten questions on sessions 1 to 4, with the answers in collapsed boxes. None of these traps is on the real quiz. Try each one before opening its answer.

Open the Practice Quiz

Reading

Lovelace, Nowosad, and Muenchow. *Geocomputation with R.* Ch. 4, Spatial data operations. (link)

Gimond, Manuel. *Intro to GIS and Spatial Analysis.* Ch. 8, Spatial Operations and Vector Overlays. (link)

Lovelace, Nowosad, and Muenchow. *Geocomputation with R.* Ch. 5, Geometry operations. (link)

5 Rasters — Continuous Surfaces and Zonal Statistics

Lecture

Open the Slides

Reading

Lovelace, Nowosad, and Muenchow. *Geocomputation with R.* Ch. 6, Raster-vector interactions. (link)

Gimond, Manuel. *Intro to GIS and Spatial Analysis.* Ch. 10, Map Algebra. (link)

Hijmans, Robert. *Spatial Data Science with R and terra.* (link)

Optional: Creating a Website with GitHub Pages and Quarto

Lecture

Open the Slides

Reading

Quarto. *Creating a Website.* (link)

Quarto. *Publishing to GitHub Pages.* (link)

GitHub. *GitHub Pages Documentation.* (link)